

MLX91804 TPMS Application Example

1. Abstract

This document describes a firmware implementing TPMS application example for the MLX91804/5 device.

2. RFTx reports

The firmware example implements periodical measurements of battery voltage, temperature, pressure and acceleration with the next RF reporting.

The default reports' periodicity is about 4s that can be changed by LF commands.

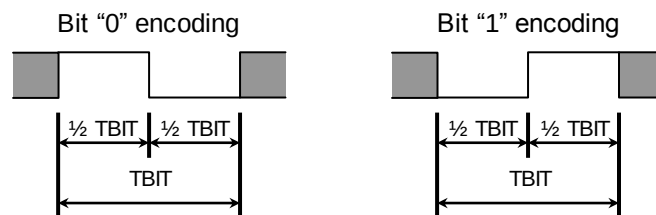
Between the RFTx telegrams and between the states when the MCU is active, the part stays in the sleep mode to save energy.

If temperature change exceeds 10°C, the RFTx calibration is done before reporting.

2.1. RF carrier frequency and data encoding

The table below specifies the low level parameters of the RF communication channel:

Carrier frequency	433,92 MHz
Modulation type	FSK programmed approximately with ± 35 kHz deviation
Data encoding type	Manchester
Raw data rate	9587 bits per second that corresponds to the standard bit rate 9600 bits per second



Manchester bits encoding used in the example with RFTX_BIT_INVERT=0

2.2. Reporting messages description

Each report includes 4 identical RF telegrams (frames) transmitted with random delays within a range 50...150 milliseconds. The number of telegrams per report is defined by constant FRAMES_per_REPORT (see "main.c").

There is a structure of the report telegram (MSB is going first):

1 byte	1 byte	4 bytes	1 byte	1 byte	2 bytes	1 byte	1 byte
Preamble	Sync_word	ID	Pressure	Temperature	Acceleration	BatteryVoltage	CRC8

Where:

Preamble is a 1 byte field with value 0x00, Manchester coded.

Sync_word is a 1 byte field with value 0x01, Manchester coded.

ID field includes 4 bytes of the unique chip ID of the MLX91804 part.

Pressure is an unsigned 8 bits value (1 byte) with pressure data.

"Pressure" byte can be converted to the [kPa] scale by using formulas below:

If "Pressure" = 0x00: measurement error happened
 If "Pressure" = 0x01: Pressure_kPa < 100 kPa (underflow)
 If "Pressure" = 0x02: Pressure_kPa = 100 kPa
 If "Pressure" = 0xFE: Pressure_kPa = 730 kPa
 If "Pressure" = 0xFF: Pressure_kPa > 730 kPa (overflow)
 Else Pressure_kPa = 100 + 2.5 * ("Pressure" - 2), expressed in [kPa]

Temperature is an unsigned 8 bits value (1 byte) with temperature data.

"Temperature" byte can be converted to the [°C] scale by using formulas below:

If "Temperature" = 0x00: measurement error happened
 If "Temperature" = 0x01: Temperature_degC < -40°C
 If "Temperature" = 0xF8: Temperature_degC > 125°C
 Else Temperature_degC = "Temperature" - 52, expressed in [°C]

Acceleration is 16-bit (2 bytes, MSB first) field with acceleration data expressed:

- in ADC output scale, if ADC_LOW_POWER_ACCELERATION option is defined;
- in "0.1 * g" otherwise.

BatteryVoltage is an unsigned 8 bits value (1 byte) with data about the actual battery voltage.

"BatteryVoltage" can be converted to the [Volt] scale by using formulas below:

If "BatteryVoltage" = 0x00: Battery_Voltage_V < 1.5 V
 If "BatteryVoltage" = 0xFF: Battery_Voltage_V > 4.0 V
 Else Battery_Voltage_V = ("BatteryVoltage" + 150.) / 100.

2.3. CRC calculation

CRC8

“CRC8” is 8-bit CRC result generated by MLX91804 HW according to the Baicheva's 8-bit polynomial = $x^8 + x^5 + x^3 + x^2 + x + 1$ (0x12F) with preload value 0xAA.

A C-function below can be helpful at the receiver side to calculate such kind of CRC:

```
/* -----
 * 8-bit CRC calculation
 * Arguments:   uint8 const *data - pointer to the buffer with data covered by CRC
 *              uint8 len        - number of bytes to be covered by CRC
 * Returns:     The computed 8-bit CRC value of the buffer's data.*/

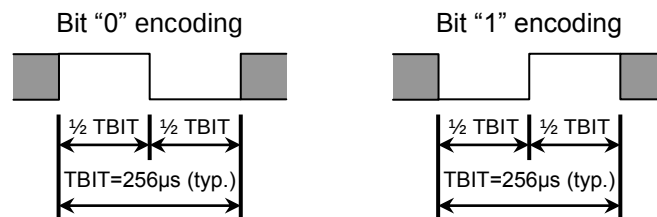
static const unsigned char CRC8_Table[256] = {
/*0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | A | B | C | D | E | F */
0x00, 0x2f, 0x5e, 0x71, 0xbc, 0x93, 0xe2, 0xcd, 0x57, 0x78, 0x09, 0x26, 0xeb, 0xc4, 0xb5, 0x9a,
0xae, 0x81, 0xf0, 0xdf, 0x12, 0x3d, 0x4c, 0x63, 0xf9, 0xd6, 0xa7, 0x88, 0x45, 0x6a, 0x1b, 0x34,
0x73, 0x5c, 0x2d, 0x02, 0xcf, 0xe0, 0x91, 0xbe, 0x24, 0x0b, 0x7a, 0x55, 0x98, 0xb7, 0xc6, 0xe9,
0xdd, 0xf2, 0x83, 0xac, 0x61, 0x4e, 0x3f, 0x10, 0x8a, 0xa5, 0xd4, 0xfb, 0x36, 0x19, 0x68, 0x47,
0xe6, 0xc9, 0xb8, 0x97, 0x5a, 0x75, 0x04, 0x2b, 0xb1, 0x9e, 0xef, 0xc0, 0xd0, 0x22, 0x53, 0x7c,
0x48, 0x67, 0x16, 0x39, 0xf4, 0xdb, 0xaa, 0x85, 0x1f, 0x30, 0x41, 0x6e, 0xa3, 0x8c, 0xfd, 0xd2,
0x95, 0xba, 0xcb, 0xe4, 0x29, 0x06, 0x77, 0x58, 0xc2, 0xed, 0x9c, 0xb3, 0x7e, 0x51, 0x20, 0x0f,
0x3b, 0x14, 0x65, 0x4a, 0x87, 0xa8, 0xd9, 0xf6, 0x6c, 0x43, 0x32, 0x1d, 0xd0, 0xff, 0x8e, 0xa1,
0xe3, 0xcc, 0xbd, 0x92, 0x5f, 0x70, 0x01, 0x2e, 0xb4, 0x9b, 0xea, 0xc5, 0x08, 0x27, 0x56, 0x79,
0x4d, 0x62, 0x13, 0x3c, 0xf1, 0xde, 0xaf, 0x80, 0x1a, 0x35, 0x44, 0x6b, 0xa6, 0x89, 0xf8, 0xd7,
0x90, 0xbf, 0xce, 0xe1, 0x2c, 0x03, 0x72, 0x5d, 0xc7, 0xe8, 0x99, 0xb6, 0x7b, 0x54, 0x25, 0x0a,
0x3e, 0x11, 0x60, 0x4f, 0x82, 0xad, 0xdc, 0xf3, 0x69, 0x46, 0x37, 0x18, 0xd5, 0xfa, 0x8b, 0xa4,
0x05, 0x2a, 0x5b, 0x74, 0xb9, 0x96, 0xe7, 0xc8, 0x52, 0x7d, 0x0c, 0x23, 0xee, 0xc1, 0xb0, 0x9f,
0xab, 0x84, 0xf5, 0xda, 0x17, 0x38, 0x49, 0x66, 0xfc, 0xd3, 0xa2, 0x8d, 0x40, 0x6f, 0x1e, 0x31,
0x76, 0x59, 0x28, 0x07, 0xca, 0xe5, 0x94, 0xbb, 0x21, 0x0e, 0x7f, 0x50, 0x9d, 0xb2, 0xc3, 0xec,
0xd8, 0xf7, 0x86, 0xa9, 0x64, 0x4b, 0x3a, 0x15, 0x8f, 0xa0, 0xd1, 0xfe, 0x33, 0x1c, 0x6d, 0x42};

unsigned char CalcCRC8(unsigned char const *data, unsigned char len)
{
    unsigned char crc = 0xAA;

    while ((len--) != 0)
    {
        crc ^= *data;
        crc = CRC8_Table[crc];
        data++;
    }
    return crc;
} /* End of CalcCRC8 */
/* -----*/
```

3. LF commands

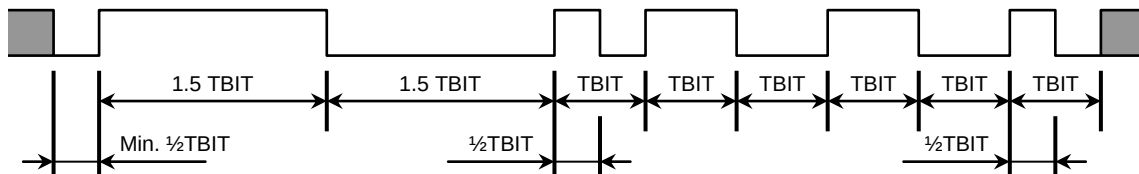
The RFTx reports' periodicity can be configured by means of the LF commands transmitted with ASK modulated 125 kHz carrier frequency. The data is coded by the inverse Manchester (biphase-L) with 50% duty cycle and raw data rate 3.9 Kbit per second:



Inverse Manchester encoding used in LF communication

There is a structure of the LF commands:

9 bits	2 bytes	4 fixed bytes	1 byte
Synchronization field (see formal below)	Wakeup_ID: 0x8F61	Command ID (just for example): 0x01, 0x02, 0x03, 0x04	Command Code



Synchronization field format

Wakeup_ID field (also known as a Header) includes 2 bytes: 0x8F (first) and 0x61 (second). One can choose another values if needed.

Command ID field includes 4 bytes with fixed values = 0x01, 0x02, 0x03, 0x04. These values mean nothing and were taken as an example.

Command Code This 8-bit field includes either 1 byte with identical nibbles that define the periodicity of RF reports, or a fixed value 0x5A to provoke a Software Reset.

This is a list of acceptable values for the field "Periodicity of RF reports":

0x00, 0x11, 0x22, 0x33, 0x44, 0x55, 0x66, 0x77, 0x88, 0x99, 0xAA, 0xBB, 0xCC, 0xDD, 0xEE, 0xFF or 0x5A.

If the Command Code equals to 0x5A, the SW reset will be generated in 8 seconds. This feature can be useful to provoke a cold bootstrap and by this way to pass execution to the LF boot loader.

Otherwise, the nibble of the Command Code specifies a report periodicity expressed in seconds. For example, the value 0x77 will set 7 seconds for the report periodicity.

Zero value disables measurements and reports until the next LF command with non-zero nibbles is received.

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